



Short Review

RIFM fragrance ingredient safety assessment, 2,6-xylenol, CAS Registry Number 576-26-1

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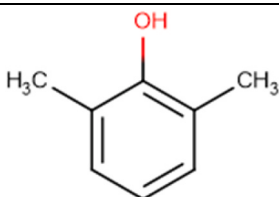
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Name: 2,6-Xylenol

CAS Registry Number: 576-26-1

Abbreviation/Definition List:

2-Box Model - A RIFM, Inc. proprietary *in silico* tool used to calculate fragrance air exposure concentration

AF - Assessment Factor

BCF - Bioconcentration Factor

CAESAR - Computer-Assisted Evaluation of industrial chemical Substances According to Regulations

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CNIH - Confirmation of No Induction in Humans test. A human repeat insult patch test that is performed to confirm an already determined safe use level for fragrance ingredients (Na et al., 2021)

Creme RIFM Model - The Creme RIFM Model uses probabilistic (Monte Carlo) simulations to allow full distributions of data sets, providing a more realistic estimate of aggregate exposure to individuals across a population (Comiskey et al., 2015, 2017; Safford et al., 2015, 2017, 2024) compared to a deterministic aggregate approach

DEREK - Derek Nexus is an *in silico* tool used to identify structural alerts

DRF - Dose Range Finding

DST - Dermal Sensitization Threshold

ECHA - European Chemicals Agency; please note that the citation dates used for studies sourced from the ECHA website are the dates the dossiers were first published, not the dates that the studies were conducted

ECOSAR - Ecological Structure-Activity Relationships Predictive Model

EU - Europe/European Union

GLP - Good Laboratory Practice

HESS - Hazard Evaluation Support System; a repeated dose profiler that is used to identify the toxicological profiler of chemicals

IFRA - The International Fragrance Association

IRB - Institutional Review Board

ISS - Istituto Superiore di Sanità (Italian National Institute of Health)

LOEL - Lowest Observed Effect Level

MOE - Margin of Exposure

MPPD - Multiple-Path Particle Dosimetry. An *in silico* model for inhaled vapors used to simulate fragrance lung deposition

NA - North America

NESIL - No Expected Sensitization Induction Level

NOAEC - No Observed Adverse Effect Concentration

NOAEL - No Observed Adverse Effect Level

NOEC - No Observed Effect Concentration

NOEL - No Observed Effect Level

OASIS - OASIS Laboratory of Mathematical Chemistry (LMC)

OECD - Organisation for Economic Co-operation and Development

OECD TG - Organisation for Economic Co-operation and Development Testing Guidelines

PBT - Persistent, Bioaccumulative, and Toxic

PEC/PNEC - Predicted Environmental Concentration/Predicted No Effect Concentration

Perfumery - In this safety assessment, perfumery refers to fragrances made by a perfumer used in consumer products only. The exposures reported in the safety assessment include consumer product use but do not include occupational exposures.

QRA - Quantitative Risk Assessment

QSAR - Quantitative Structure-Activity Relationship

REACH - Registration, Evaluation, Authorisation, and Restriction of Chemicals

RfD - Reference Dose

RIFM - Research Institute for Fragrance Materials

RQ - Risk Quotient

Statistically Significant - Statistically significant difference in reported results as compared to controls with a $p < 0.05$ using appropriate statistical test

Toxtree - an *in silico* tool that can estimate toxic hazard by applying a decision tree approach

TTC - Threshold of Toxicological Concern

UV/Vis spectra - Ultraviolet/Visible spectra

VCF - Volatile Compounds in Food

VoU - Volume of Use

vPvB - (very) Persistent, (very) Bioaccumulative

WoE - Weight of Evidence

The Expert Panel for Fragrance Safety* concludes that this material is safe as described in this safety assessment.

This safety assessment is based on the RIFM Criteria Document (Api, 2015), which should be referred to for clarifications.

Each endpoint discussed in this safety assessment includes the relevant data that were available at the time of writing (version number in the top box is indicative of the date of approval based on a 2-digit month/day/year), both in the RIFM Database (consisting of publicly available and proprietary data) and through publicly available information sources (e.g., SciFinder and PubMed). Studies selected for this safety assessment were based on appropriate test criteria, such as acceptable guidelines, sample size, study duration, route of exposure, relevant animal species, most relevant testing endpoints, etc. A key study for each endpoint was selected based on the most conservative endpoint value (e.g., PNEC, NOAEL, LOEL, and NESIL).

*The Expert Panel for Fragrance Safety is an independent body that selects its own members and establishes its own operating procedures. The Expert Panel is

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comprised of internationally known scientists that provide RIFM with guidance relevant to human health and environmental protection.

Summary: The existing information supports the use of this material as described in this safety assessment.

2,6-Xylenol was evaluated for genotoxicity, repeated dose toxicity, reproductive toxicity, local respiratory toxicity, photoirritation/photoallergenicity, skin sensitization, and environmental safety. Data show that 2,6-xylenol is not genotoxic. Data on 2,6-xylenol provide a calculated Margin of Exposure (MOE) > 100 for the repeated dose toxicity, developmental toxicity, and local respiratory toxicity endpoints. Data on read-across analog xyleneol (CAS # 1300-71-6) provide a calculated MOE > 100 for the fertility endpoint. The skin sensitization endpoint was completed using the Dermal Sensitization Threshold (DST) for reactive materials ($64 \mu\text{g}/\text{cm}^2$); exposure is below the DST. The photoirritation/photoallergenicity endpoints were evaluated based on ultraviolet/visible (UV/Vis) spectra; 2,6-xylenol is not expected to be photoirritating/photoallergenic. The environmental endpoints were evaluated; 2,6-xylenol was found not to be Persistent, Bioaccumulative, and Toxic (PBT) as per the International Fragrance Association (IFRA) Environmental Standards, and its risk quotients, based on its current volume of use (VoU) in Europe and North America (i.e., Predicted Environmental Concentration/Predicted No Effect Concentration [PEC/PNEC]), are < 1 .

Human Health Safety Assessment

Genotoxicity: Not genotoxic. ECHA (2010)

Repeated Dose Toxicity: NOAEL = 33 mg/kg/day. (BGChemie, 2005)

Reproductive Toxicity: Developmental NOAEL = 180 mg/kg/day; Fertility NOAEL = 245 mg/kg/day. (ECHA, 2010; ECHA, 2016)

Skin Sensitization: Not a concern for skin sensitization under the declared use levels; exposure is below the DST.

Photoirritation/Photoallergenicity: Not expected to be photoirritating/photoallergenic. (UV/Vis spectra, RIFM Database)

Local Respiratory Toxicity: NOAEC = 200 mg/m³. ECHA (2010)

Environmental Safety Assessment

Hazard Assessment:

Persistence: Screening-level: 2.83 (BIOWIN 3) (EPI Suite v4.11; US EPA, 2012a)

Bioaccumulation: Screening-level: 16.75 L/kg (EPI Suite v4.11; US EPA, 2012a)

Ecotoxicity: Screening-level: Fish LC50: 48.55 mg/L (RIFM Framework; Salvito, 2002)

Conclusion: Not PBT or vPvB as per IFRA Environmental Standards

Risk Assessment:

Screening-level: PEC/PNEC (North America and Europe) < 1 (RIFM Framework; Salvito, 2002)

Critical Ecotoxicity Endpoint: Fish LC50: 48.55 mg/L (RIFM Framework; Salvito, 2002)

RIFM PNEC is: 0.04855 $\mu\text{g}/\text{L}$

• **Revised PEC/PNECs (2019 IFRA VoU):** North America and Europe: Not applicable; cleared at the screening-level

1. Identification

- Chemical Name:** 2,6-Xylenol
- CAS Registry Number:** 576-26-1
- Synonyms:** 2,6-Dimethylphenol; 2-Hydroxy-1,3-dimethylbenzene; Phenol, 2,6-dimethyl-; 1-Hydroxy-2,6-dimethylbenzene; 2,6-Xylenol
- Molecular Formula:** C₈H₁₀O
- Molecular Weight:** 122.16 g/mol
- RIFM Number:** 6153
- Stereochemistry:** No stereocenter present and no stereoisomer possible.

2. Physical data

- Boiling Point:** 203 °C (Fragrance Materials Association [FMA]), 210.67 °C (EPI Suite v4.11)
- Flash Point:** 86 °C (Globally Harmonized System)
- Log K_{ow}:** 2.36 (measured); 2.37 (calculated) (Smith et al., 2002), 2.61 (EPI Suite v4.11)

- 4. **Melting Point:** 45 °C (FMA), 25.37 °C (EPI Suite v4.11)
- 5. **Water Solubility:** 3615 mg/L at 25 °C (EPI Suite v4.11)
- 6. **Specific Gravity:** Not Available
- 7. **Vapor Pressure:** 0.1 mm Hg at 20 °C (FMA), 0.105 mm Hg (EPI Suite v4.11)
- 8. **UV Spectra:** No absorbance between 290 and 700 nm; molar absorption coefficient is below the benchmark (1000 L mol⁻¹ • cm⁻¹)
- 9. **Appearance/Organoleptic:** Sweet, "rooty" odor (Arctander, Volume II, 1969)

3. Volume of use (worldwide band)

- 1. 0.1–1 metric ton per year (IFRA, 2019)

4. Exposure to fragrance ingredient (Creme RIFM aggregate exposure model v3.3.10)

- 1. **95th Percentile Concentration in Fine Fragrance:** 0.0004% (RIFM, 2023)
- 2. **Inhalation Exposure*:** 0.000067 mg/kg/day or 0.00046 mg/day (RIFM, 2023)
- 3. **Total Systemic Exposure**:** 0.00013 mg/kg/day (RIFM, 2023)

*95th percentile calculated exposure derived from concentration survey data in the Creme RIFM Aggregate Exposure Model (Comiskey et al., 2015, 2017; Safford et al., 2015, 2017, 2024).

**95th percentile calculated exposure; assumes 100% absorption unless modified by dermal absorption data as reported in Section V. It is derived from concentration survey data in the Creme RIFM Aggregate Exposure Model and includes exposure via dermal, oral, and inhalation routes whenever the fragrance ingredient is used in products that include these routes of exposure (Comiskey et al., 2015, 2017; Safford et al., 2015, 2017, 2024).

5. Derivation of systemic absorption

- 1. **Dermal:** Assumed 100%
- 2. **Oral:** Assumed 100%
- 3. **Inhalation:** Assumed 100%

6. Computational toxicology evaluation

6.1. Cramer classification: class II, intermediate* (expert judgment)

Expert Judgment	Toxtree v3.1	OECD QSAR Toolbox v4.6 (OECD, 2023)
II	I	I

*See the Appendix below for details.

6.2. Analogs selected

- a. Genotoxicity: None
- b. Repeated Dose Toxicity: None
- c. Reproductive Toxicity: Mixture of xylene isomers (CAS # 1300-71-6)
- d. Skin Sensitization: None
- e. Photoirritation/Photoallergenicity: None
- f. Local Respiratory Toxicity: None
- g. Environmental Toxicity: None
- 3. Read-across Justification: See Appendix below

7. Metabolism

No relevant data are available for inclusion in this safety assessment.
Additional References: None.

8. Natural occurrence

2,6-Xylenol is reported to occur in the following foods by the VCF*.

Cheese, various types	Lamb's lettuce (<i>Valerianella locusta</i>)
Coffee	Pork
Fish	Rum
Honey	Salami
Katsuoibushi (dried bonito)	Whiskey

*VCF (Volatile Compounds in Food): Database/Nijssen, L.M.; Ingen-Visscher, C.A. van; Donders, J.J.H. (eds). – Version 15.1 – Zeist (The Netherlands): TNO Triskelion, 1963–2014. A continually updated database containing information on published volatile compounds that have been found in natural (processed) food products. Includes FEMA GRAS and EU-Flavis data.

9. Reach dossier

Available (ECHA, 2010); accessed on 03/07/24.

10. Conclusion

The existing information supports the use of this material as described in this safety assessment.

11. Summary

11.1. Human health endpoint summaries

11.1.1. Genotoxicity

Based on the current existing data, 2,6-xyleneol does not present a concern for genotoxicity.

11.1.1.1. Risk assessment. The mutagenic activity of 2,6-xyleneol has been evaluated in a bacterial reverse mutation assay conducted in compliance with GLP regulations and in accordance with OECD TG 471 using the standard plate incorporation and preincubation methods. *Salmonella typhimurium* strains TA98, TA100, TA1535, and TA1537 were treated with 2,6-xyleneol in ethanol at concentrations up to 5000 µg/plate. No increases in the mean number of revertant colonies were observed at any tested concentration in the presence or absence of S9 (ECHA, 2010). Under the conditions of the study, 2,6-xyleneol was not mutagenic in the Ames test.

The clastogenicity of 2,6-xyleneol was assessed in an *in vitro* chromosome aberration study conducted in compliance with GLP regulations and in accordance with OECD TG 473. Chinese hamster lung cells (V79) were treated with 2,6-xyleneol in ethanol at concentrations up to 1220 µg/mL in the dose range finding (DRF) study; the main study was conducted at concentrations up to 600 µg/mL in the presence or absence of S9. Statistically significant increases in the frequency of cells with structural chromosomal aberrations or polyploid cells were observed in the presence of S9 after treatment with 2,6-xyleneol in the first study at the 28-h preparation interval at 600 µg/mL and in the second study at the 18-h preparation interval at 400 µg/mL and the 28-h preparation interval at 300 µg/mL (ECHA, 2010). Under the conditions of the study, 2,6-xyleneol was considered to be clastogenic in the *in vitro* chromosome aberration assay with metabolic activation.

To further investigate the results of the *in vitro* chromosome aberration study, the clastogenicity of 2,6-xyleneol was assessed in an *in vivo* chromosome aberration study conducted in compliance with GLP regulations and in an equivalent manner to OECD TG 475. The test material was administered in corn oil via oral gavage to groups of male and female Sprague Dawley rats at doses up to 3500 mg/kg body weight in the DRF study. Doses of 350, 700, and 1400 mg/kg in males and doses of 300, 600, and 1200 mg/kg in females were administered in the

definitive assay. Rats from each dose level were euthanized at 12, 24, and 36 h, and the bone marrow was extracted. No statistically significant increases in the frequency of cells with structural chromosomal aberrations or polyploid cells were observed with any dose of the test material (ECHA, 2010). Under the conditions of the study, 2,6-xylenol was considered to be non-clastogenic in the *in vivo* chromosome aberration assay.

Additional References: Epler et al., 1979

Literature Search and Risk Assessment Completed On: 04/26/24.

11.1.2. Repeated dose toxicity

The MOE for 2,6-xylenol is adequate for the repeated dose toxicity endpoint at the current level of use.

11.1.2.1. Risk assessment. There are sufficient repeated dose toxicity data on 2,6-xylenol. In an OECD 407 gavage 28-day subchronic toxicity study, groups of 5 Wistar rats/sex/dose were administered 2,6-xylenol at doses of 0, 20, 100, 400, or 800 mg/kg/day 2,6-xylenol in an olive oil vehicle for 5 days/week for 4 weeks. Observations included clinical signs, body weight, feed consumption, hematology, clinical chemistry, organ weights, gross necropsy, and histopathology. One rat at 800 mg/kg/day died after the third dose, showing perforation of the forestomach, inflammatory edema of the submucosa, and purulent peritonitis at necropsy. At 800 mg/kg/day, reduced bodyweight gain, increased water consumption, and adverse clinical signs (hypothermia, ataxia, salivation, reduced general state) were noted. In females at 800 mg/kg/day, anemia was observed (reduced red blood cells, hemoglobin, and hematocrit). In both sexes, at 800 mg/kg/day, liver weights were increased, but there were corresponding histopathological changes. Erosion and ulceration of the glandular stomach were noted in half the high-dose group, and increased extramedullary hematopoiesis was observed in 8/10 rats. At 400 mg/kg/day, increased water consumption, adverse clinical signs (ataxia, salivation, reduced general state) in both sexes, and anemia in the females were noted. Similarly to the high dose, liver weights were increased, and extramedullary hematopoiesis was observed in the spleen. At 100 mg/kg/day, liver weights were increased in females. No effects were observed at 20 mg/kg/day. The NOEL in males was determined to be 100 mg/kg/day for males and 20 mg/kg/day for females. The NOAEL was determined to be 100 mg/kg/day (BGChemie, 2005).

A default safety factor of 3 was used when deriving a NOAEL from OECD 407 studies (ECHA, 2012). The safety factor has been approved by the Expert Panel for Fragrance Safety*.

The derived NOAEL for the repeated dose toxicity endpoint is 100/3 or 33 mg/kg/day.

Therefore, the 2,6-xylenol MOE for the repeated dose toxicity endpoint can be calculated by dividing the 2,6-xylenol NOAEL in mg/kg/day by the total systemic exposure to 2,6-xylenol, 33/0.00013 or 253846.

Additionally, the total systemic exposure to 2,6-xylenol (0.13 µg/kg/day) is below the TTC (9 µg/kg/day; Kroes et al., 2007) for the repeated dose toxicity endpoint of a Cramer Class II material at the current level of use.

*The Expert Panel for Fragrance Safety is composed of scientific and technical experts in their respective fields. This group provides advice and guidance.

Additional References: None.

Literature Search and Risk Assessment Completed On: 04/26/24.

11.1.3. Reproductive toxicity

The MOE for 2,6-xylenol is adequate for the reproductive toxicity endpoint at the current level of use.

11.1.3.1. Risk assessment. The developmental toxicity data on 2,6-xylenol are sufficient for the developmental toxicity endpoint. An OECD 414 gavage developmental toxicity study was conducted in CD Sprague Dawley rats. Groups of 24 pregnant rats were gavaged with 0, 60, 180, or 540 mg/kg/day 2,6-xylenol on gestation days 6–15. Observations included mortality, clinical signs, body weight, and feed consumption. On day 20, dams were euthanized and Caesarean sectioned, livers were weighed, and a complete macroscopic examination was conducted. Ovaries and uterine contents were evaluated for weight, number of corpora lutea, implantations, early and late resorptions, and live and dead fetuses. Each fetus was weighed, and sexes were examined externally for malformations/variations and examined for soft tissue and head or skeletal malformations/variations. At 540 mg/kg/day, mortality (2/24), bodyweight loss, reduced weight gain, and decreased food consumption were observed. Pups from these rats had reduced fetal body weights. At 180 mg/kg/day, maternal body weight (during gestation) and weight gain were reduced. The NOAEL for maternal toxicity was determined to be 60 mg/kg/day. The NOAEL for developmental toxicity was determined to be 180 mg/kg/day (ECHA, 2010).

Therefore, the 2,6-xylenol MOE for the developmental toxicity endpoint can be calculated by dividing the 2,6-xylenol NOAEL in mg/kg/day by the total systemic exposure to 2,6-xylenol, 180/0.00013 or 1384615.

There are no fertility data on 3,4-xylenol. However, a mixture of xylenol isomers (CAS # 1300-71-6; see Section VI) containing 16.4% 2,5-xylenol (CAS # 95-87-4), 16.9% 3,4-xylenol (CAS # 95-65-8), 22.7% 2,4-xylenol (CAS # 105-67-9), 11.1% 3,5-dimethylphenol (CAS # 108-68-9), 18.2% 2,3-dimethylphenol (CAS # 526-75-0), and 14.7% 2,6-xylenol (CAS # 576-26-1) has sufficient data to support the fertility endpoint.

In a GLP- and OECD 422-compliant study, groups of 10 CrI:CD (SD) IGS BR VAF/Plus rats/sex/dose were administered xylenol (isomer mixture) via oral gavage at doses of 0, 30, 100, or 245 mg/kg/day for 28 days in males and 54 days in females. There were no mortalities throughout the study. There were clinical signs, including urine-stained fur at the high dose in both sexes. Relative ovary weight was higher in females at the high dose. Decreased mating frequency was observed (i.e., 8/10 rats mated at the high dose versus 10/10 in the control), but this was not statistically significant and was within the historical control range. Therefore, this was not considered related to the treatment. Thus, based on no adverse effects up to the highest dose, the reproductive toxicity NOAEL for this study was determined to be 245 mg/kg/day (ECHA, 2016).

Therefore, the 2,6-xylenol MOE for the fertility endpoint can be calculated by dividing the xylenol (isomer mixture) NOAEL in mg/kg/day by the total systemic exposure to 2,6-xylenol, 245/0.00013 or 1884615.

Additionally, the total systemic exposure to 2,6-xylenol (0.13 µg/kg/day) is below the TTC (9 µg/kg/day; Kroes et al., 2007; Laufersweiler et al., 2012) for the reproductive toxicity endpoint of a Cramer Class II material at the current level of use.

Additional References: None.

Literature Search and Risk Assessment Completed On: 04/26/24.

11.1.4. Skin sensitization

Based on existing data and the application of DST, 2,6-xylenol does not present a safety concern for skin sensitization under the current, declared levels of use.

11.1.4.1. Risk assessment. No skin sensitization data are available for 2,6-xylenol (Table 1). 2,6-Xylenol is predicted *in silico* to be non-reactive with skin proteins directly (Roberts et al., 2007; Toxtree v3.1.0; OECD Toolbox v4.6). Acting conservatively due to the limited data, the reported exposure was benchmarked utilizing the reactive DST of 64

Table 1
Summary of existing data on 2,6-xyleneol.

WoE Skin Sensitization Potency Category ¹	Human Data				Animal Data		
	NOEL-CNIH (induction) $\mu\text{g}/\text{cm}^2$	NOEL-HMT (induction) $\mu\text{g}/\text{cm}^2$	LOEL (induction) $\mu\text{g}/\text{cm}^2$	WoE NESIL $\mu\text{g}/\text{cm}^2$	LLNA Weighted Mean EC3 Value $\mu\text{g}/\text{cm}^2$	GPMT	Buehler
Human potency category unknown; Current exposure level below the DST for reactive materials.	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	In vitro Data				In silico protein binding alerts (OECD Toolbox v4.6)		
	KE 1	KE 2	KE 3	Target Material	Autoxidation simulator	Metabolism simulator	
	N/A	N/A	N/A	No alert found	Michael addition; Radical reactions	No alert found	

NOEL = No observed effect level; CNIH = Confirmation of No Induction in Humans; HMT = Human Maximization Test; LOEL = lowest observed effect level; EC3 = concentration of test chemical required to induce a 3-fold increase in lymph node cell proliferation; GPMT = Guinea Pig Maximization Test; KE = Key Event; N/A = Not Available.
¹WoE Skin Sensitization Potency Category is only applicable for identified sensitizers with sufficient data, based on collective consideration of all available data (Na et al., 2021).

$\mu\text{g}/\text{cm}^2$ (Safford, 2008; Safford, 2011; Roberts et al., 2015; Safford, 2015). The current exposure from the 95th percentile concentration is below the DST for reactive materials when evaluated in all QRA categories. Table 2 provides the supported concentrations for 2,6-xyleneol that present no appreciable risk for skin sensitization based on the reactive DST. These levels represent supported concentrations based on the DST approach. However, additional studies may show it could be used at higher levels.

Additional References: Yamano et al., 2007
Literature Search and Risk Assessment Completed On: 04/23/24.

11.1.5. Photoirritation/photoallergenicity

Based on the available UV/Vis absorption spectra, 2,6-xyleneol would not be expected to present a concern for photoirritation or photoallergenicity.

11.1.5.1. Risk assessment. There are no photosafety studies available for 2,6-xyleneol in experimental models. UV/Vis absorption spectra indicate no absorption between 290 and 700 nm. Thus, it does not present a concern for photoirritant or photoallergenic effects (Henry, 2009). Based on the lack of absorbance, 2,6-xyleneol does not present a

concern for photoirritation or photoallergenicity.

11.1.5.2. UV spectra analysis. UV/Vis absorption spectra (OECD TG 101) were obtained. The spectra indicate no absorbance in the range of 290–700 nm. Thus, it does not present a concern for photoirritant or photoallergenic effects (Henry, 2009).

Additional References: None.
Literature Search and Risk Assessment Completed On: 04/24/24.

11.1.6. Local respiratory toxicity

The MOE for 2,6-xyleneol is adequate for the respiratory endpoint at the current level of use.

11.1.6.1. Risk assessment. The calculated chronic inhalation exposure was considered along with toxicological data observed in the scientific literature to calculate the MOE from inhalation exposure when used in perfumery. In a 2-week acute study conducted in rats, a NOAEC of 200 mg/m^3 was reported for 2,6-xyleneol (ECHA, 2010). Ten Fischer rats per sex per dose were treated with 2,6-xyleneol at concentrations 0, 67, 200, and 670 mg/m^3 for 6 h a day and 5 days per week for 2 weeks. Test substance-related microscopic findings were noted to be confined in the

Table 2

Supported concentrations for 2,6-xylenol that present no appreciable risk for skin sensitization based on reactive DST.

IFRA Category ^a	Description of Product Type	Supported Concentrations ^b (%) in Finished Products Based on Reactive DST	Reported 95th Percentile Use Concentrations in Finished Products
1	Products applied to the lips	0.0049	NRU ^d
2	Products applied to the axillae	0.0015	2.2 x 10 ⁻⁵
3	Products applied to the face using fingertips	0.029	NRU ^d
4	Fine fragrance products	0.027	4.0 x 10 ⁻⁴
5	Products applied to the face and body using the hands (palms), primarily leave-on	0.0070	4.4 x 10 ⁻⁶
6	Products with oral and lip exposure	0.016	0.0015
7	Products applied to the hair with some hand contact	0.056	4.3 x 10 ⁻⁶
8	Products with significant anogenital exposure	0.0029	No Data ^c
9	Products with body and hand exposure, primarily rinse-off	0.054	6.6 x 10 ⁻⁵
10	Household care products with mostly hand contact	0.19	9.1 x 10 ⁻⁴
11	Products with intended skin contact but minimal transfer of fragrance to skin from inert substrate	0.11	No Data ^c
12	Products not intended for direct skin contact, minimal or insignificant transfer to skin	Not restricted	1.4

^a For a description of the categories, refer to the IFRA/RIFM Information Booklet.

^b These levels represent maximum acceptable concentrations based on the DST. However, additional studies may show it could be used at higher levels.

^c Fragrance exposure from these products is very low. These products are not currently in the Creme RIFM Aggregate Exposure Model.

^d No reported use.

nasal cavity of high-dose male and female rats. The lesions were observed in levels II, III, and IV, with the most extensive effects showing at level IV. The effects ranged from mild to severe in proportion to the amount of involved inflammatory, necrotic, and degenerative lesions. Microscopic observations revealed necrosis/degeneration of the olfactory epithelium, which was the most severely affected area in the dorsal meatus and nasal septum. The overall thickness of the affected regions of olfactory epithelium showed a slight change in the less affected regions, to a gradual thinning into a single layer of cuboidal cells, and finally to complete denudation in the most severely affected areas involving the dorsal meatus. The submucosa in the dorsal meatus was thickened and presented with inflammatory exudates, edema fluid, and proliferating loose fibrous connective tissue. Occasionally, the olfactory nerve fibers appeared to be vacuolated in appearance in areas of severe inflammation. However, it could not be confirmed whether these olfactory nerve effects were fixation artifacts or treatment-related neural degeneration. Focal sub-pleural inflammatory aggregates were observed in the male

and female animals from treated and control groups and were not considered treatment-related. There were no effects seen in the animals from low- and mid-dose groups for the upper respiratory regions, and other than the pulmonary inflammatory foci, no effects were seen in the lower respiratory regions. Therefore, the NOAEC was identified to be 200 mg/m³.

This NOAEC expressed in mg/kg lung weight/day is.

- $(200 \text{ mg/m}^3) \times (1 \text{ m}^3/1000 \text{ L}) = 0.2 \text{ mg/L}$
- Minute ventilation of 0.17 L/min (Arms and Travis, 1988) for a rat $\times$ duration of exposure of 360 min per day (min/day) (according to GLP study guidelines) = 61.2 L/day
- $(0.2 \text{ mg/L}) \times (61.2 \text{ L/d}) = 12.24 \text{ mg/day}$
- $(12.24 \text{ mg/day}) / (0.0016 \text{ kg lung weight of rat [Phalen, 2009]}) = 7650 \text{ mg/kg lung weight/day}$

The 95th percentile calculated exposure was reported to be 0.00046 mg/day—this value was derived from the concentration survey data in the Creme RIFM exposure model (Comiskey et al., 2015; B. Safford et al., 2015). To compare this estimated exposure with the NOAEC expressed in mg/kg lung weight/day, this value is divided by 0.65 kg human lung weight (Carthew, 2009) to give 0.0007 mg/kg lung weight/day resulting in a MOE of 10,928,571 (i.e., $[7650 \text{ mg/kg lung weight of rat/day}] / [0.0007 \text{ mg/kg lung weight of human/day}]$).

The MOE is greater than 100. Without adjustment for specific uncertainty factors related to inter-species and intra-species variation, for study in humans, the material exposure by inhalation at 0.00046 mg/day is deemed to be safe under the most conservative consumer exposure scenario.

Additional References: Uzhdavini et al., 1970

Literature Search and Risk Assessment Completed On: 04/16/24.

11.2. Environmental endpoint summary

11.2.1. Screening-level assessment

A screening-level risk assessment of 2,6-xylenol was performed following the RIFM Environmental Framework (Salvito, 2002), which provides 3 tiered levels of screening for aquatic risk. In Tier 1, only the material's regional VoU, its log K_{OW}, and its molecular weight are needed to estimate a conservative risk quotient (RQ), expressed as the ratio Predicted Environmental Concentration/Predicted No Effect Concentration (PEC/PNEC). A general QSAR with a high uncertainty factor applied is used to predict fish toxicity, as discussed in Salvito et al. (2002). In Tier 2, the RQ is refined by applying a lower uncertainty factor to the PNEC using the ECOSAR model (US EPA, 2012b), which provides chemical class-specific ecotoxicity estimates. Finally, if necessary, Tier 3 is conducted using measured biodegradation and ecotoxicity data to refine the RQ, thus allowing for lower PNEC uncertainty factors. The data for calculating the PEC and PNEC for this safety assessment are provided in the table below. For the PEC, the range from the most recent IFRA VoU Survey is reviewed. The PEC is then calculated using the actual regional tonnage, not the extremes of the range. Following the RIFM Environmental Framework, 2,6-xylenol was identified as a fragrance material with no potential to present a possible risk to the aquatic environment (i.e., its screening-level PEC/PNEC <1).

A screening-level hazard assessment using EPI Suite v4.11 (US EPA, 2012a) did not identify 2,6-xylenol as possibly being persistent or bioaccumulative based on its structure and physical-chemical properties. This screening-level hazard assessment considers the potential for a material to be persistent and bioaccumulative and toxic, or very persistent and very bioaccumulative as defined in the Criteria Document (Api, 2015). As noted in the Criteria Document, the screening criteria applied are the same as those used in the EU for REACH (ECHA, 2017a). For persistence, if the EPI Suite model BIOWIN 3 predicts a value < 2.2 and either BIOWIN 2 or BIOWIN 6 predicts a value < 0.5, then the

	LC50 (Fish)	EC50 (<i>Daphnia</i>)	EC50 (Algae)	AF	PNEC	Chemical Class
RIFM Framework Screening-level (Tier 1)	<u>48.55</u>			1000000	0.04855	

material is considered potentially persistent. A material would be considered potentially bioaccumulative if the EPI Suite model BCFBAF predicts a fish BCF ≥ 2000 L/kg. Ecotoxicity is determined in the above screening-level risk assessment. If, based on these model outputs (Step 1), additional assessment is required, a WoE-based review is then performed (Step 2). This review considers available data on the material's physical-chemical properties, environmental fate (e.g., OECD Guideline biodegradation studies or die-away studies), fish bioaccumulation, and higher-tier model outputs (e.g., US EPA's BIOWIN and BCFBAF found in EPI Suite v4.11). Data on persistence and bioaccumulation are reported below and summarized in the Environmental Safety Assessment section prior to Section I.

11.2.1.1. Risk assessment. Based on the current VoU (IFRA, 2019), 2,6-xyleneol does not present a risk to the aquatic compartment in the screening-level assessment.

11.2.1.2. Key studies

11.2.1.2.1. Biodegradation. No data available.

11.2.1.2.2. Ecotoxicity. Phipps et al., 1981: The 96-h acute toxicity study was conducted with Fathead minnows according to the OECD 203 method under flow-through conditions. The 96-h LC50 for the test material was >27 mg/L.

11.2.1.2.3. Other available data. 2,6-Xyleneol has been registered under REACH, and the following data is available (ECHA, 2010):

A 72-h algae growth inhibition test was conducted according to the OECD 201 method. The EC50 was reported to be 10 mg/L and 45 mg/L for the area under the growth curve and growth rate, respectively.

A *Daphnia magna* reproduction test was conducted according to the OECD 211 method. The 21-day No Observed Effect Concentration (reproduction) was reported to be 0.54 mg/L.

11.2.1.3. Risk assessment refinement. Since 2,6-xyleneol passed the screening-level (Tier 1), the measured data are included for completeness only and have not been used in PNEC derivation.

Ecotoxicological data and PNEC derivation (all endpoints reported in mg/L; PNECs in $\mu\text{g/L}$)

Endpoints used to calculate PNEC are underlined.

Exposure information and PEC calculation (following RIFM Environmental Framework: Salvito et al., 2002)

Exposure	Europe (EU)	North America (NA)
Log K_{ow} Used	2.61	2.61
Biodegradation Factor Used	0	0
Dilution Factor	3	3
Regional VoU Tonnage Band	<1	<1
Risk Characterization: PEC/PNEC	<1	<1

Based on available data, the RQ for this material is < 1 . No further assessment is necessary.

The RIFM PNEC is 0.04855 $\mu\text{g/L}$. The revised PEC/PNECs for EU and NA are not applicable. The material was cleared at the screening-level; therefore, it does not present a risk to the aquatic environment at the current reported VoU.

Literature Search and Risk Assessment Completed On: 04/08/24.

Literature search*

- **RIFM Database:** Target, Fragrance Structure-Activity Group materials, other references, JECFA, CIR, SIDS
- **ECHA:** <https://echa.europa.eu/>
- **NTP:** <https://ntp.niehs.nih.gov/>
- **OECD Toolbox:** <https://www.oecd.org/chemicalsafety/risk-assessment/oecd-qsar-toolbox.htm>
- **SciFinder:** <https://scifinder.cas.org/scifinder/view/scifinder/scifinderExplore.jsf>
- **PubChem:** <https://pubchem.ncbi.nlm.nih.gov/>
- **PubMed:** <https://www.ncbi.nlm.nih.gov/pubmed>
- **National Library of Medicine Technical Bulletin:** https://www.nlm.nih.gov/pubs/techbull/nd19/nd19_toxnet_new_locations.html
- **IARC:** <https://monographs.iarc.fr>
- **OECD SIDS:** <https://hpvchemicals.oecd.org/ui/Default.aspx>
- **EPA ACToR:** <https://actor.epa.gov/actor/home.xhtml>
- **US EPA ChemView:** <https://chemview.epa.gov/chemview/>
- **Japanese NITE:** https://www.nite.go.jp/en/chem/chrip/chrip_search/systemTop
- **Japan Existing Chemical Data Base (JECDB):** http://dra4.nihs.go.jp/mhlw_data/jsp/SearchPageENG.jsp
- **Google:** <https://www.google.com>
- **ChemIDplus:** <https://pubchem.ncbi.nlm.nih.gov/source/ChemIDplus>

Search keywords: CAS number and/or material names.

*Information sources outside of RIFM's database are noted as appropriate in the safety assessment. This is not an exhaustive list. The links listed above were active as of 10/31/24.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper. We wish to confirm that there are no known conflicts of interest associated with this publication and there has been no significant financial support for this work that could have influenced its outcome. RIFM staff are employees of the Research Institute for Fragrance Materials, Inc. (RIFM). The Expert Panel receives a small honorarium for time spent reviewing the subject work.

Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.fct.2024.115118>.

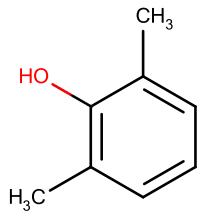
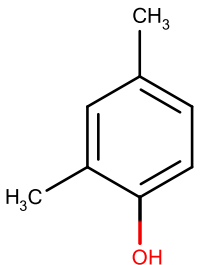
Appendix

Read-across Justification:

Methods

The read-across analog was identified using RIFM fragrance chemicals inventory clustering and read-across search criteria (Date et al., 2020). These criteria are in compliance with the strategy for structuring and reporting a read-across prediction of toxicity as described in Schultz et al. (2015) and are consistent with the guidance provided by OECD within Integrated Approaches for Testing and Assessment (OECD, 2015) and the European Chemicals Agency read-across assessment framework (ECHA, 2017b).

- First, materials were clustered based on their structural similarity. Second, data availability and data quality on the selected cluster were examined. Third, appropriate read-across analogs from the cluster were confirmed by expert judgment.
- Tanimoto structure similarity scores were calculated using FCFC4 fingerprints (Rogers and Hahn, 2010).
- The physical–chemical properties of the target material and the read-across analogs were calculated using EPI Suite (US EPA, 2012a).
- $J_{\max}$ values were calculated using RIFM's skin absorption model (SAM). The parameters were calculated using the consensus model (Shen et al., 2014).
- DNA binding, mutagenicity, genotoxicity alerts, and oncologic classification predictions were generated using OECD QSAR Toolbox v4.6 (OECD, 2023).
- ER binding and repeat dose categorization were generated using OECD QSAR Toolbox v4.6 (OECD, 2023).
- Developmental toxicity was predicted using CAESAR v2.1.7 (Cassano et al., 2010), and skin sensitization was predicted using Toxtree v2.6.13.
- Protein binding was predicted using OECD QSAR Toolbox v4.6 (OECD, 2023).
- The major metabolites for the target material and read-across analogs were determined and evaluated using OECD QSAR Toolbox v4.6 (OECD, 2023).
- To keep continuity and compatibility with *in silico* alerts, OECD QSAR Toolbox v4.6 was selected as the alert system.

	Target Material	Read-across Material
Principal Name	2,6-Xylenol	Xylenol
CAS No.	576-26-1	1300-71-6
Structure		
Similarity (Tanimoto Score)		0.93
SMILES	Cc1cccc(C)c1O	Cc1ccc(O)c(C)c1
Endpoint		Fertility
Molecular Formula	C ₈ H ₁₀ O	C ₈ H ₁₀ O
Molecular Weight (g/mol)	122.167	122.167
Melting Point (°C, EPI Suite)	45.70	24.50
Boiling Point (°C, EPI Suite)	201.00	210.90
Vapor Pressure (Pa @ 25°C, EPI Suite)	2.28E+01	1.36E+01
Water Solubility (mg/L, @ 25°C, WSKOW v1.42 in EPI Suite)	6.05E+03	7.87E+03
Log K _{ow}	2.36	2.3
$J_{\max}$ (µg/cm ² /h, SAM)	420.20	507.10
Henry's Law (Pa·m ³ /mol, Bond Method, EPI Suite)	6.74E-01	9.64E-02
Reproductive Toxicity		
ER Binding (OECD QSAR Toolbox v4.6)	Non-binder, impaired OH or NH ₂ group	Weak binder, OH group
Developmental Toxicity (CAESAR v2.1.6)	Toxicant (low reliability)	Toxicant (low reliability)
Metabolism		
Rat Liver S9 Metabolism Simulator and Structural Alerts for Metabolites (OECD QSAR Toolbox v4.6)	See Supplemental Data 1	See Supplemental Data 2

Summary

There are insufficient toxicity data on 2,6-xylenol (CAS # 576-26-1). Hence, *in silico* evaluation was conducted to determine read-across analogs for this material. Based on structural similarity, reactivity, physical–chemical properties, and expert judgment, xylenol (CAS # 1300-71-6) was identified

as a read-across analog with sufficient data for toxicological evaluation.

Conclusions

- Xylenol (CAS # 1300-71-6) was used as a read-across analog for the target material, 2,6-xylenol (CAS # 576-26-1), for the fertility endpoint.
 - o The target material and the read-across analog are structurally similar and belong to the group of alkyl-substituted phenols.
 - o The key difference between the target material and the read-across analog is that the read-across analog is substituted in the *ortho* and *para* positions, while the target material is substituted only in the *ortho* position. The read-across analog contains the structural features of the target material that are relevant to this endpoint and is expected to have equal or greater potential for toxicity as compared to the target material.
 - o The similarity between the target material and the read-across analog is indicated by the Tanimoto score. Differences between the structures that affect the Tanimoto score are toxicologically insignificant.
 - o The physical–chemical properties of the target material and the read-across analog are sufficiently similar to enable a comparison of their toxicological properties.
 - o According to the OECD QSAR Toolbox v4.6, structural alerts for toxicological endpoints are consistent between the target material and the read-across analog.
 - o Both the target material and read-across analog contain *in silico* alerts for non-toxicants. The read-across analog contains an alert for weak binders, whereas the target material contains an alert for non-binders (developmental toxicity and fertility). The data from the fertility section confirms that the MOE of the target material is adequate under the current usage. The predictions are superseded by the data.
 - o The target material and the read-across analog are expected to be metabolized similarly, as shown by the metabolism simulator.
 - o The structural alerts for the endpoints evaluated are consistent between the metabolites of the read-across analog and the target material.

Explanation of Cramer Classification:

Due to potential discrepancies with the current *in silico* tools (Bhatia et al., 2015), the Cramer class of the target material was determined using expert judgment based on the Cramer decision tree (Cramer et al., 1978).

- Q1. Normal constituent of the body? No
- Q2. Contains functional groups associated with enhanced toxicity? No
- Q3. Contains elements other than C, H, O, N, divalent S? No
- Q5. Simply branched aliphatic hydrocarbon or a common carbohydrate? No
- Q6. Benzene derivative with certain substituents? No
- Q7. Heterocyclic? No
- Q16. Common terpene? No
- Q17. Readily hydrolyzed to a common terpene? No
- Q19. Open chain? No
- Q23. Aromatic? Yes
- Q27. Rings with substituents? Yes
- Q28. More than one aromatic ring? No
- Q30. Aromatic Ring with complex substituents? No
- Q18. One of the list (see Cramer et al., 1978 for a detailed explanation on the list of categories)? Yes, Class Intermediate (Class II)

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